

Ensemble method:

↳ combine multiple model (weak learners) to create a single, more powerful model predictive model

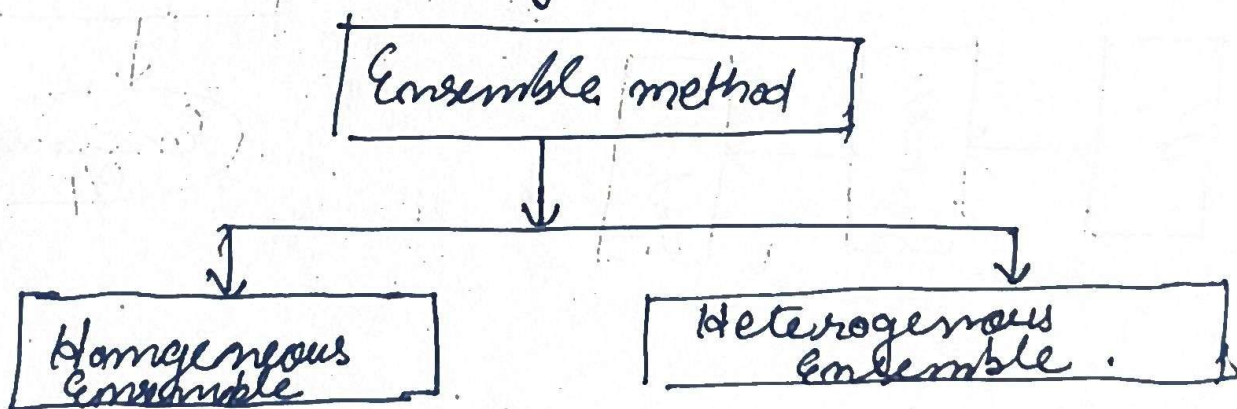
why use Ensemble method:

↳ Reduce of overfitting: combine model smooths out individual mistakes and stabilize variance

↳ Improve accuracy: The aggregated ensemble usually outperforms its best individual component

↳ Handle noise: They are inherently more strong to data outliers and noisy training sets.

→ Thus, main objective of ensemble method is decrease the Bias (underfitting) and variance (overfitting) and improve the overall accuracy.

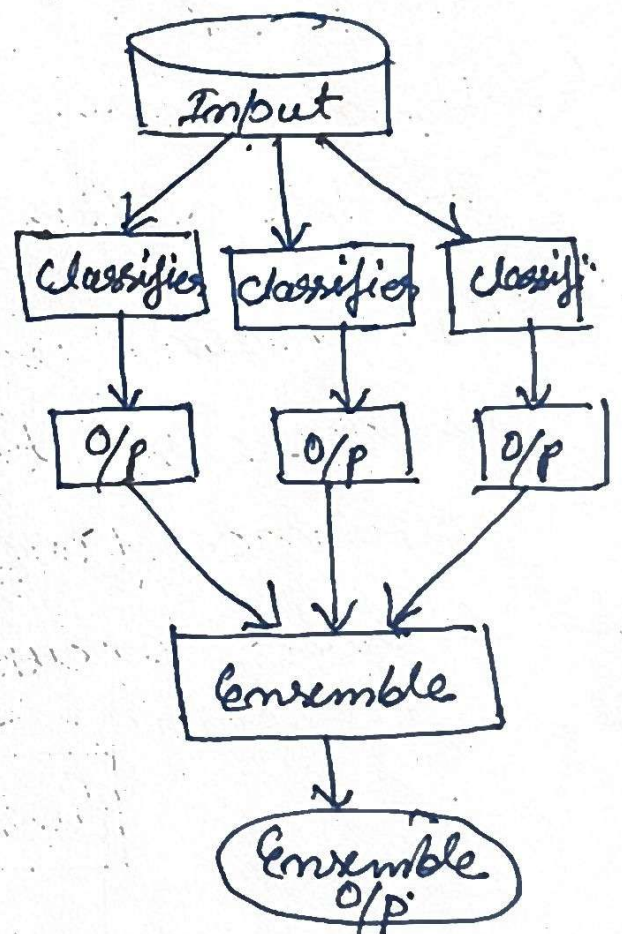
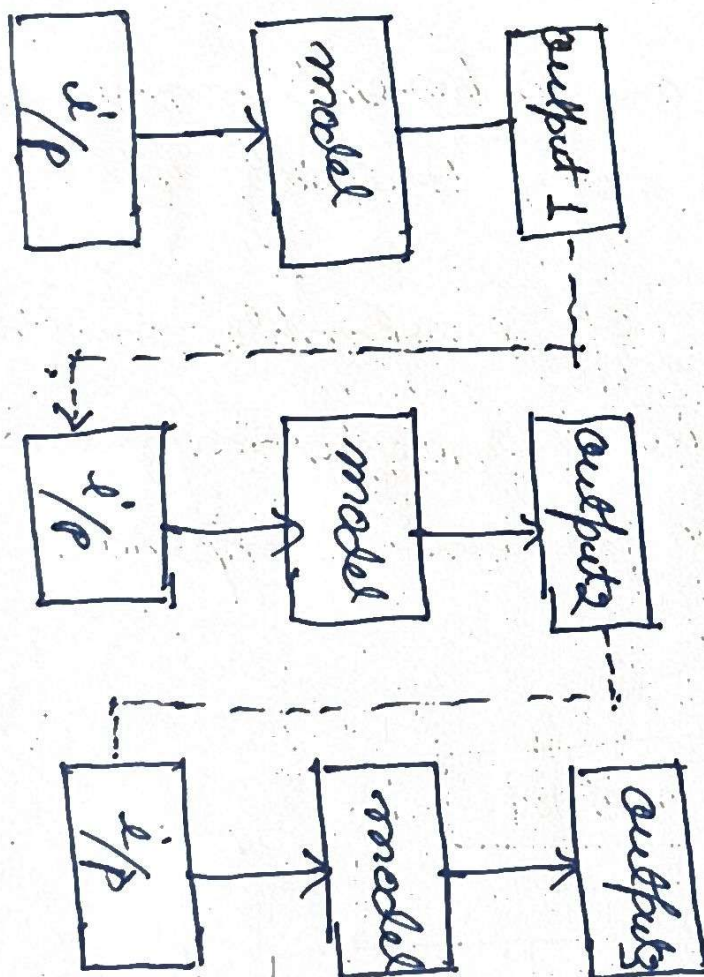
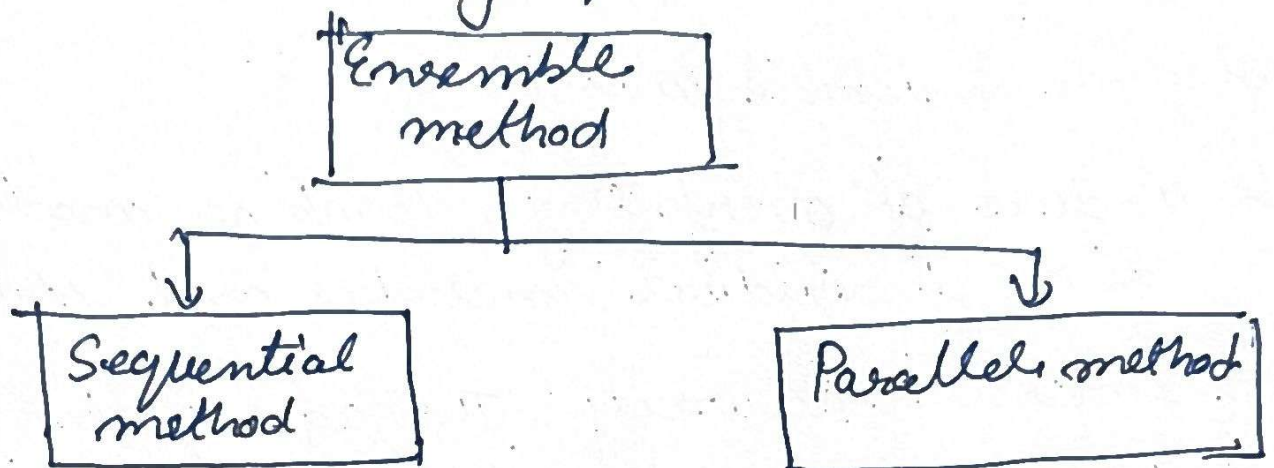


all base learners are same

Base learner are different type.

# Fuzzy Set operations:

Based on learning approach



Ensemble method:

↳ Ensemble methods are machine learning techniques that combine multiple models to improve accuracy and stability.

↳ Ensemble methods are ideal for regression and classification, where they reduce bias and variance to boost the accuracy.

↳ The most popular ensemble methods are boosting, bagging and stacking.

↳ Ensemble methods fall into two broad categories

• ↳ Sequential Ensemble

↳ Parallel Ensemble

↳ Majority of ensemble techniques apply a single algorithm in base learning, which results in homogenous base learners.

↳ Heterogenous base learners are learners of distinct types.

Sequential ensemble techniques.

↳ It generates base learner in a sequence

Ex AdaBoost (Adaptive Boosting), Gradient Boosting (GB)

↳ The sequential generation of base learners promotes the dependence between the base learners

↳ The performance of the model is then improved by assigning higher weights to previously misclassified learners.

Parallel ensemble techniques:

↳ Base learners are generated in a parallel format Ex random forest

↳ It <sup>uses</sup> ~~utilize~~ the parallel generation of base learner to encourage independence between the base learners

↳ The independence of base learners significantly reduces the error due to application of average.

## Main Type of Ensemble Methods.

↳ Bagging

↳ Boosting

↳ Stacking

### Bagging

↳ Bagging is the short form of bootstrap aggregating

↳ It mainly applied in classification and regression.

↳ It involves training multiple models independently and combining their predictions through averaging or voting.

↳ A dataset is randomly selected with replacement to construct several subsets. Known as bootstrap samples in bagging

↳ A basic model, such as a decision tree or random forest is trained separately on each bootstrap sample.

↳ Each model's output is then aggregated

often by taking the average for regression or the mode for classification problems.

↳ Bagging may be used with a number of different base models, including decision trees, neural networks, support vector machines.

↳ Bagging is especially successful when the base models have high variance and large quantity of training data is available.

↳ There are <sup>five</sup> ~~four~~ stages in bagging algorithm.

↳ ~~1. Bagging method~~

↳ 1. Bootstrap Sampling

(Process of randomly sampling a dataset with replacement to generate various subsets k/a bootstrap samples)

↳ 2. Base Model Training

(A base model like DT, ANN, SVM is trained individually on each bootstrap sample. Subsets are not similar, each base model generates a separate prediction model)

↳ 3. Aggregation

(The result of each model is then aggregated)

# Ensemble Learning

Voting

- Same data & different model

Bagging

Pasting

- Bagging without replacement

Stacking

- Blender
- No aggregation

Adaptive Boosting

- weight increases for misclassification and weights decreases for right classification

Boosting

Light GBM

Cat GBM

Extreme Gradient Boosting (XGB)

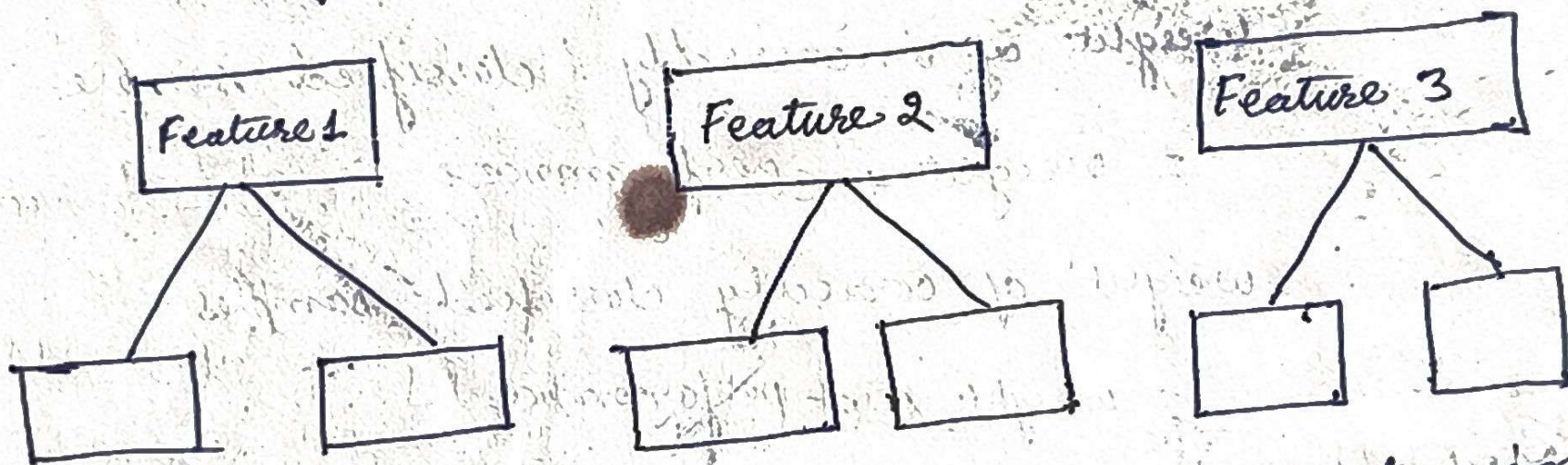
Gradient Boosting (GB)

- Instead of weights, here gradient (residuals, loss) is passed in next model.

## AdaBoost:

### Algorithm steps.

- 1) Initially, all the samples will have the same weight  $w = \frac{1}{N}$
- 2) create first base learner i.e. the first stump



- 3) AdaBoost will calculate the total error by using the formula

$$\frac{\text{No. of samples classified as wrong}}{\text{Total No. of samples}}$$

Ex if error = 2, sample = 5

$$\text{Total Error (TE)} = \frac{2}{5}$$

#### Step 4

After calculating the total error now, AdaBoost will calculate the performance of stamp

$$\text{Performance of stamp} = \frac{1}{2} \ln \left[ \frac{1 - TE}{TE} \right]$$

#### Step 5:

Now, AdaBoost will update the weights of correctly and incorrectly classified samples.

weight of incorrectly classified sample

$$= \text{weight} * e^{\text{performance}}$$

weight of correctly classified samples

$$= \text{weight} * e^{-\text{performance}}$$

#### Step 6:

Now, in this last step, the AdaBoost will create a new dataset for the next weak learner

In this step, high probability for the wrong records to get selected several time.

→ New model will be  $f_t = f_{t-1} + \lambda_t g_t$

→ output the final model  $f_T(x) = \text{sgn} \left( \sum_{t=20}^T \lambda_t g_t \right)$

# Adaboost Algorithm

Given:  $(x_1, y_1), \dots, (x_m, y_m)$  where  $x_i \in \mathcal{X}$ ,  $y_i \in \{-1, +1\}$ .

Initialize:  $D_1(i) = 1/m$  for  $i = 1, \dots, m$ .

For  $t = 1, \dots, T$ :

- Train weak learner using distribution  $D_t$ .
- Get weak hypothesis  $h_t : \mathcal{X} \rightarrow \{-1, +1\}$ .
- Aim: select  $h_t$  with low weighted error:

$$\varepsilon_t = \Pr_{i \sim D_t} [h_t(x_i) \neq y_i].$$

- Choose  $\alpha_t = \frac{1}{2} \ln \left( \frac{1 - \varepsilon_t}{\varepsilon_t} \right)$ .
- Update, for  $i = 1, \dots, m$ :

$$D_{t+1}(i) = \frac{D_t(i) \exp(-\alpha_t y_i h_t(x_i))}{Z_t}$$

where  $Z_t$  is a normalization factor (chosen so that  $D_{t+1}$  will be a distribution).

Output the final hypothesis:

$$H(x) = \text{sign} \left( \sum_{t=1}^T \alpha_t h_t(x) \right).$$